Slic3r: A Reliable 3D Printing Companion

The use of 3D printing has become extensively popular over the last few years. Software has taken up a gamut of responsibilities to aid the process. In this article we explore one such useful software called Slic3r—a 3D slicing software for 3D printers, available to users as a free open source entity.

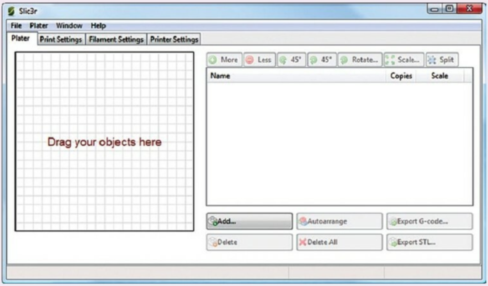

Fig. 1: Slic3r interface on start up (Credit: slic3r.org)

Familiarising with Slic3r

Built using C++ 11 and Python (for graphical interface), Slic3r is a toolpath generator for 3D printers based on digital prototypes. This means that Slic3r generates G-codes from created 3D digital models, which can be used as instructions by the 3D printer to generate the physical object.

Slic3r works on all major operating systems, namely, Mac OSX, Windows and Linux. It has support for all major G-code languages including Marlin, Repetier, Mach3, Makerware, Smoothie, Sailfish, LinuxCNC and Machinekit. The motive of this software is to feed in existing 3D model files, generate G-codes from these and, eventually, print the object from a connected 3D printer.

Slic3r's interface is compact, clutter-free and easy to use. At the top, there is the general menu bar containing File, Plater, Window and Help tabs. These can be used for various operations like loading existing files and more. Below this are four essential tabs that allow you to set up the 3D model files and the printer to ensure proper printing of the model. These tabs are: Plater for configuring the digital model and setting it for print, Print Settings for choosing various printing preferences, Filament Settings for proper arrangement of the print filament and Printer Settings for configuring the preferred printer with the system to develop the model.

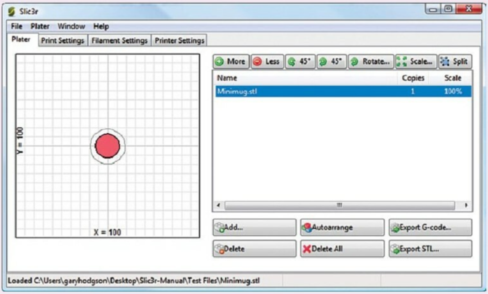

Fig. 2: Plater tab (Credit: slic3r.org)

The main workspace is split in two. Under Plater, the left side holds the main visual representation of the file. It is a graphical bed interface with clearly-specified X and Y axes. The right panel consists of the list of imported files and a number of actionable buttons to make limited modifications to the file. Opening an existing file can be either done by clicking Add or by simply dragging-and-dropping. You can add multiple files, all of which are listed in the